

Motion Planning Evaluation Tests: Repeatability and Obstacle Handling

MoveIt Task Constructor on the xArm 7, E-Waste Battery Extraction, Redback Operations

Prepared by Tarunjeet Singh, Project Lead

1. Purpose

This document extends the earlier planner comparison with two structured tests of the same MoveIt Task Constructor (MTC) pipeline on the xArm 7 [1], [2]. Test 1 repeats planning ten times per planner in the open scene to measure repeatability. Test 2 places a wall between the battery and the drop zone to measure how each planner handles an obstacle. The results are used to refine the planner recommendation before hardware validation.

2. Test Setup

All runs used the xarm7_mtc_planners package, the same ten stage pick and place task and the same scene as the earlier case documents. Runs were planned without execution, so every run started from the same home configuration, and each run appended one row to a results file.

Setting	Value
Planners	OMPL (PipelinePlanner), Cartesian (CartesianPath), joint interpolation (JointInterpolationPlanner)
Task	Above battery, descend, attach, lift, transfer, lower, detach, retreat, home
Battery	70 x 50 x 10 mm at (0.40, 0.00, 0.005) m in link_base
Drop zone	120 x 120 x 5 mm at (0.30, -0.30, 0.0025) m
Obstacle (Test 2 only)	Wall 120 x 20 x 300 mm at (0.35, -0.15, 0.15) m, across the straight line from battery to drop zone
Execution	Disabled for batch runs (execute:=false)
Batch command	run_batch.sh 10 false (Test 1) and run_batch.sh 10 true (Test 2)
Platform	Ubuntu 24.04 in WSL2, ROS 2 Jazzy, MoveIt 2, MTC Jazzy, ros2_control fake hardware

Planning time is the wall clock time of task.plan(), including planning scene retrieval.

Execution time is the summed trajectory duration. Joint path length is the summed Euclidean distance between consecutive joint space waypoints. Values are reported as mean $\pm$ standard deviation over successful runs.

3. Test 1: Repeatability in the Open Scene

Planner	Success	Plan (ms)	Exec (s)	Joint path (rad)
OMPL	10/10	531.3 $\pm$ 38.3	14.378 $\pm$ 0.003	5.925 $\pm$ 0.002
Cartesian	10/10	394.7 $\pm$ 37.9	14.647 $\pm$ 0.000	5.972 $\pm$ 0.000
Joint interpolation	10/10	352.7 $\pm$ 61.2	14.381 $\pm$ 0.000	5.926 $\pm$ 0.000

- **Determinism confirmed:** Cartesian and joint interpolation returned an identical path on all ten runs, with zero variation in cost, execution time and joint path length.
- **OMPL is near deterministic in open space:** with no obstacle forcing a detour, OMPL converged to almost the same joint path as joint interpolation, with a standard deviation of only 0.002 rad.
- **Planning time:** joint interpolation planned fastest, followed by Cartesian and then OMPL. This corrects the single run result in the earlier comparison, where Cartesian appeared fastest.
- **Motion quality:** Cartesian produced the longest motion, about 0.8 percent more joint travel and 1.9 percent more execution time, because it holds the tool on straight lines.

4. Test 2: Obstacle Scene

The results file holds 24 OMPL runs and 20 runs each for Cartesian and joint interpolation, all recorded with the wall in place.

Planner	Success	Plan (ms)	Exec (s)	Joint path (rad)
OMPL	14/24 (58%)	584.9 ± 56.7	25.58 ± 7.12	11.57 ± 3.78
Cartesian	0/20 (0%)	-	-	-
Joint interpolation	0/20 (0%)	-	-	-

Failed runs took 528.3 ± 49.0 ms for OMPL, 375.4 ± 68.4 ms for Cartesian and 372.7 ± 29.8 ms for joint interpolation before reporting failure.

The effect of the wall on OMPL is shown against its open scene result:

OMPL metric	Open scene	Obstacle scene	Change
Success rate	100%	58%	-42 points
Joint path (rad)	5.925 ± 0.002	11.57 ± 3.78	1.95 times longer
Joint path range (rad)	5.923 to 5.928	8.17 to 23.20	Wide spread
Execution time (s)	14.378	25.58 (19.45 to 46.82)	1.78 times longer
Planning time (ms)	531.3	584.9	+10%

5. Failure Analysis

One diagnostic run per planner was repeated with the wall in place to capture the MTC stage report. OMPL was run four times and succeeded once.

Planner	Failing stage	MTC message	Interpretation
Cartesian	Transfer to drop zone	CartesianPath: min_fraction not met. Achieved: 0.218750	The straight line reached 21.9% of its length before the wall blocked it
Joint interpolation	Transfer to drop zone	Waypoint is in collision!	The direct joint space path passes through the wall

Planner	Failing stage	MTC message	Interpretation
OMPL	Return home	No solution (0/1)	Transfer succeeds, but the path from the drop zone back to home is not found in 3 of 4 diagnostic runs

- **Cartesian and joint interpolation:** both planners solved every stage up to the transfer. Neither searches for an alternative route, so a single blocking obstacle makes the whole task infeasible.
- **OMPL:** in all three diagnostic failures, OMPL had already planned past the wall during the transfer and failed at the final homing stage, where the wall also lies between the drop zone and the home configuration. Because the failures are intermittent and occur within roughly half a second, the most likely cause is that the sampling search does not find a valid path within its per stage planning budget. This is an inference from the timing and has not yet been confirmed by changing the budget.

6. Discussion

- **Obstacle avoidance is the deciding factor:** in open space the three planners produce motions within 1 percent of each other. With a single obstacle, only OMPL can complete the task.
- **OMPL trades path quality for flexibility:** its detours roughly doubled joint travel and varied widely between runs, which makes execution time less predictable.
- **OMPL reliability needs work:** a 58 percent success rate is not acceptable for hardware, where a failed plan stops the extraction cycle.
- **Planning time is not the constraint:** the largest difference between planners was about 250 ms, compared with 14 to 26 s of execution, so reliability and path quality should drive planner choice.

7. Recommendation

The tests support a hybrid pipeline in which each stage uses the planner best suited to it. MTC allows a different solver per stage without changing the task structure.

Stage	Recommended planner	Evidence
Move above battery	OMPL	Free space motion that must remain valid if the cell contains obstacles
Descend and lift	Cartesian	Straight vertical tool motion; in the Cartesian obstacle run only the transfer failed
Transfer to drop zone	OMPL	Only planner that passed the wall (Test 2)
Lower and retreat	Cartesian	Controlled straight line placement above the drop zone

Stage	Recommended planner	Evidence
Return home	OMPL with a larger planning budget	Joint interpolation fails when the direct path is blocked, and OMPL homing failed intermittently with default settings

Before hardware validation on the xArm 7, the following conditions are recommended:

- Raise the OMPL planning budget for free space stages and repeat Test 2, targeting at least 95 percent success over 20 runs.
- Add path simplification or a cost based choice among the five MTC solutions to reduce the spread in OMPL path length.
- Reserve joint interpolation for motions through space that is verified clear, such as a fixed homing corridor.
- Implement the hybrid pipeline as a fourth case and evaluate it with the same open scene and obstacle tests.

8. Limitations

- All results come from simulation with ros2_control fake hardware on WSL2; timings will differ on native Linux and on the physical arm.
- Only one obstacle geometry was tested. Obstacles above the battery or around the drop zone may change the results for the vertical stages.
- The failing stage was captured for the diagnostic runs only, not for every batch run, and the cause of the OMPL homing failures is inferred and requires confirmation.
- The battery pose is fixed; detected poses from the Computer Vision sub-team have not yet been integrated.

9. References

- [1] MoveIt Contributors, "Pick and place with MoveIt Task Constructor," MoveIt 2 Documentation. [Online]. Available: https://moveit.picknik.ai/main/doc/tutorials/pick_and_place_with_moveit_task_constructor/pick_and_place_with_moveit_task_constructor.html. Accessed: Sep. 28, 2026.
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